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Advanced Robotic Kinematic Term Project

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1 Introduction

A manipulator is defined as kinematically redundant if it has more joints variable than required taskspace variable.This kinematic redundancy leads to joint motion in the null space of the Jacobian Matrix that doesnt impact the end-effector position.

$x = f(q)$ $x \in \mathbb{R}^m$ with $n > m$
$f : \mathbb{R}^n \to \mathbb{R}^m$

Differantiating Forward Kinematic yiels;

$\dot{x} = J\dot{q}$

$\dot{x} = J\dot{q}$ is solvable when $\dot{x}$ is $\in C(J)$[Column Space of J]

To find complete solution to $\dot{x} = J\dot{q}$

1)$\dot{q}_{particular} : \dot{x} = J\dot{q}_{particular}$

2)$\dot{q}_{nullspace} : 0 = J\dot{q}_{nullspace}$

Complete Solution

$J \cdot (\dot{q}_{particular} + \dot{q}_{nullspace}) = \dot{x}$

If J is full rank, then there is unique solution for this equation.

From velocity kinematics

$\dot{q} = J^{+}\dot{x} + (I_n - J^{+}J) \cdot \dot{q}$ where $J^{+} = J^{T}(J \cdot J^{T})^{-1}$
By using this property of kinematically redundant manipulator , it can be given subtask controller to mapipulator so as to use kinematic redundancy.

In literature[1], there are several subtask.

- Singularity Avoidance
- Joint Limit Avoidance
- Impact Forces
- Potential Energy

In this project , we will deal with Impact Forces and perform this subtask controller in Matlab/Simulink.

2 Dynamic Model of Robot

In this project 3-DOF RRR manipulator is used and the model is shown in Fig.1

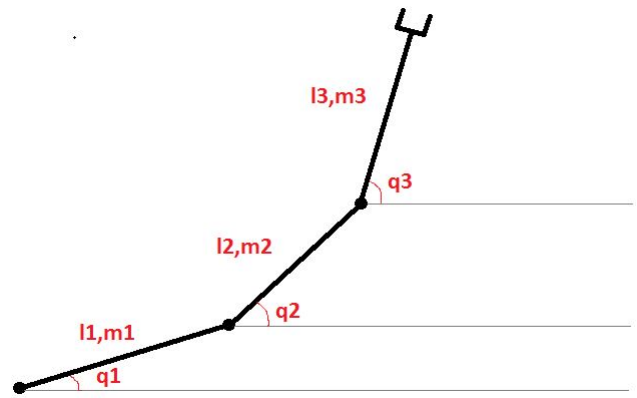

Figure 1: RRR Manipulator

Kinematic variables and equations are listed below;

	Link1	Link2	Link3
Length	0.4m	0.36m	0.3m
Mass	3.6kg	2.6kg	2kg

Friction $f_d = [5.3, 2.4, 1.1]$ and $p = [0.4752, 0.128, 0.1152]$ and $Be = [1.1956, 0.3946, 0.0512]$

$$M(q) \cdot \ddot{q} + V_m(q, \dot{q}) \cdot \dot{q} + G(q) + F_d \cdot \dot{q} = \tau \tag{1}$$

where $M(q), V_m(q, \dot{q}), G(q), F_d$ and τ are Intertia Matrix , Centriperal-Coriolis Matrix and Gravity Matrix ,Friction Matrix and Torque Vector ,respectively.

2.1 Inertia Matix

$$\text{M}=\begin{bmatrix} 2cos(q_2)p_1 + cos(q_2+q_3)p_2 + cos(q_3)p_3 + B_1 & cos(q_2)p_1 + cos(q_2+q_3)p_2 + B_2 & B_3 \\ cos(q_2)p_1 + cos(q_2+q_3)p_2 + B_2 & B_2 + 2p_3cos(q_3) & B_3 + p_3cos(q_3) \\ B_3 & B_3 + p_3cos(q_3) & B_3 \end{bmatrix}$$

2.2 Centripetal-Coriolis Matrix V_m

.

$$vm11 := -\left(\sin(q1+q2)\,p_2 + \sin(q2)\,p_1\right) dq2 - \left(\sin(q1+q2)\,p_2 + \sin(q1+q2)\,p_3\right) dq3$$

$$vm12 := -\left(\sin(q1+q2)\,p_2 + \sin(q2)\,p_1\right)(dq1+dq2) - \left(\sin(q1+q2)\,p_2 + \sin(q1+q2)\,p_3\right) dq3$$

$$vm13 := \left(\sin(q1+q2)\,p_2 + \sin(q1+q2)\,p_3\right)(-dq1+dq2+dq3)$$

$$vm21 := \left(\sin(q1+q2)\,p_2 + \sin(q2)\,p_1\right) dq1 + p_3 \sin(q3)\, dq3$$

$$vm22 := -\left(\sin(q1+q2)\,p_2 + \sin(q1+q2)\,p_3\right) dq3$$

$$vm23 := -p_3 \sin(q3)\,(3\,dq1 + dq2 + dq3)$$

$$vm31 := \left(\sin(q1+q2)\,p_2 + \sin(q2)\,p_1\right) dq1 - p_3 \sin(q3)\, dq2$$

$$vm32 := p_3 \sin(q3)\,(dq1+dq2)$$

$$vm33 := 0$$

$$V_m := \begin{bmatrix} vm11 & vm12 & vm13 \\ vm21 & vm22 & vm23 \\ vm31 & vm32 & vm33 \end{bmatrix}$$

2.3 Gravity

$$\begin{bmatrix} g(m_1 0.5 l_1 c_1 + m_2 l_1 c_1 + m_2 0.5 l_2 c_{12} + m_3 l_2 c_{12} + m_3 0.5 l_3 c_{123})g \\ (m_2 0.5 l_2 c_{12} + m_3 l_2 c_{12} + m_3 0.5 c_{123})g \\ (m_3 0.5 l_3 c_{123})g \end{bmatrix}$$

2.4 Friction Matrix

$$\begin{bmatrix} fd_1 & 0 & 0 \\ 0 & fd_2 & 0 \\ 0 & 0 & fd_3 \end{bmatrix}$$

2.5 Jacobian Matrix

$$J = \begin{bmatrix} -l_1 s_1 - l_2 s_{12} - l_3 s_{123} & -l_2 s_{12} - l_3 s_{123} & -l_3 s_{123} \\ l_1 c_1 + l_2 c_{12} + l_3 c_{123} & +l_2 c_{12} + l_3 c_{123} & l_3 c_{123} \end{bmatrix}$$

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3 Control Design

In previoius section , dynamic model of 3R redundant manipulator is given.Now we will focus on control part.

End effector tracking error is defined like

$$\dot{e} = \dot{x_d} - \dot{x} \tag{2}$$

$= \dot{x_d} - J\dot{q}$
$= K_e e - K_e e + \dot{x_d} - J\dot{q}$
$-K_e e + JJ^+ K_e e + JJ^+ \dot{x_d} - J\dot{q}$
$-K_e e + J(J^+ K_e e + J^+ \dot{x_d} + (I_n - J^+ J)g - \dot{q})$

Define

$$r = J^+(K_e e + \dot{x_d}) + (I_n - J^+ J)g - \dot{q} \tag{3}$$

$\dot{e} = -K_e e + Jr$
$\dot{r} = \frac{d}{dt}(J^+(K_e e + \dot{x_d}) + (I_n - J^+ J)g) - \ddot{q}$
$M\dot{r} = M\frac{d}{dt}(J^+(K_e e + \dot{x_d}) + (I_n - J^+ J)g) - M\ddot{q}$
$= M\frac{d}{dt}(J^+(K_e e + \dot{x_d} + (I_n - J^+ J)) + V_m(q, \dot{q})\dot{q} + G(q) + F_d\dot{q} - \tau$

Recall from definition of $r(t)$ that

$\dot{q} = -r + J^+(K_e e + \dot{x_d}) + (I_n + J^+ J)g$
$= M\frac{d}{dt}(J^+(K_e e + \dot{x_d} + (I_n - J^+ J)) + V_m(q, \dot{q})\dot{q} + G(q) + F_d\dot{q} - \tau$

$M\dot{r} = M\frac{d}{dt}(J^+(K_e e + \dot{x_d}) + (I_n - J^+ J)g) - V_m r + +V_m(J^+(K_e e + \dot{x_d}) + (I_n - J^+ J)g) + G(q) + F_d\dot{q} - \tau$

Define

$$\tau = M\frac{d}{dt}[J^+(K_e e + \dot{x_d}) + (I_n + J^+ J)g] + V_m[J^+(K_e e + \dot{x_d} + (I_n - J^+ J)g)] + G(q) + F_d\dot{q} + K_r r \tag{4}$$

where $K_r \in R^{n \times m}$ positive diagonal control gain matrix

3.1 SubTask

Define
$y_a(q) = exp(-\gamma \cdot \beta(q))$ where $\gamma \in R$ positive constant

$\beta(q)$ is designated for using subtask and $\beta(q) > 0$ is a function specific to a sub-task

As it can be seen that $0 < y_a < 1$ and as $\beta(q)$ increases y_a decreases

$$\dot{y_a} = J_s(q)\dot{q} \tag{5}$$

$J_s(q) = \frac{\partial y_a}{\partial q} \in \mathbb{R}^{1 \times n}$

$J_s = [\frac{\partial y_a}{\partial q_1}, \frac{\partial y_a}{\partial q_2}, \frac{\partial y_a}{\partial q_3}]$

Define

$$g = -k_g[J_s(I_n - J^+J)]^T y_a \tag{6}$$

where k_g positive gain

4 Beta Function $\beta(q)$

4.1 Withstanding Impacts

An alternate impact sub-task is to push the robot manipulator into postures that are best suited to witstand impacts with the environment. For this case , let $\beta(q)$ be defined as the dynamic impact measure given by the following definition

$$\beta = \sqrt{\det[(J^+)^T M^2 J^+]} \tag{7}$$

4.2 Generalized Imppact Measure

$$\beta = \sqrt{\det[(J^+)^T I^2 J^+]} \tag{8}$$

5 Simulation Results

Matlab R2017a/Simulink is used for simulations
• Solver: ODE3 (Bogacki-Shampine)
• Step Time : 0.01sec
• Stop Time : 10 sec

Simulation is designed generally as shown in the figure2 , Dynamic Model equations are given in section 2 and Control Model Equations are given in Section 3

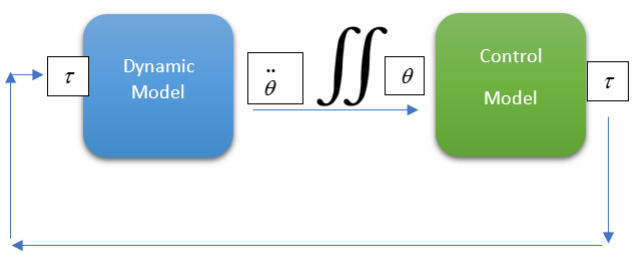


Figure 2: Design

The used trajectory equation:

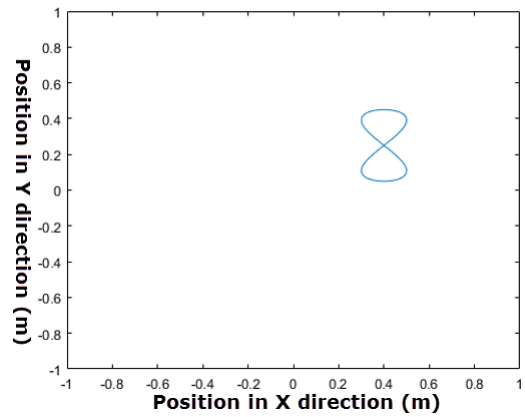


Figure 3: Desired Trajectory

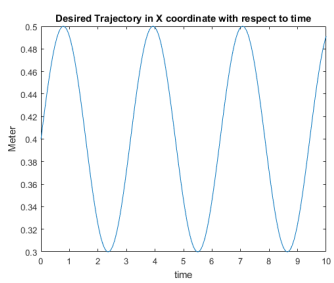
$x_d = \begin{bmatrix} 0.4 + 0.1sin(2t) \\ 0.25 + 0.2cos(t) \end{bmatrix}$

In Simulation , controler gains are choosen like this;

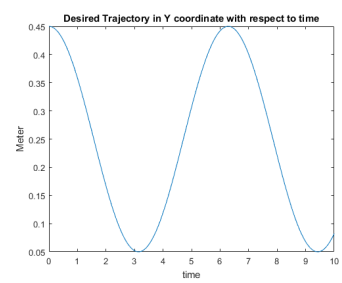
$K_r = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$ $K_e = \begin{bmatrix} 55 & 0 \\ 0 & 55 \end{bmatrix}$ $\gamma = 1$ $k_g = 1$

Initial position of robot $\theta_1 = \frac{\pi}{2}$ $\theta_2 = -\frac{\pi}{3}$ $\theta_3 = -\frac{\pi}{3}$

Desired position of end-effector of redundant robot is shown



(a) Desired Trajectory in X coordinate



(b) Desired Trajectory in Y coordinate

Figure 4: Desired Trajectory

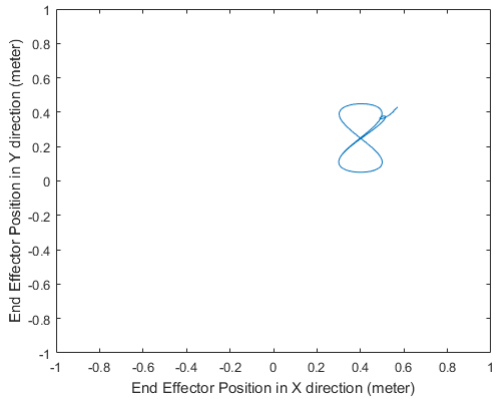
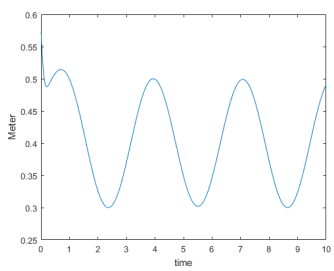
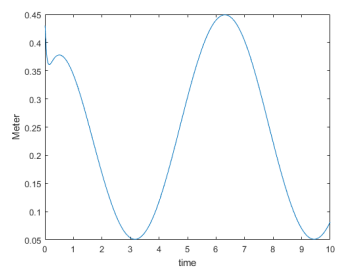


Figure 5: End Effector following trajectory



(a) End Effector following trajectory in X coordinate



(b) End Effector following trajectory in Y corrdinate

Figure 6: End Effector following trajectory

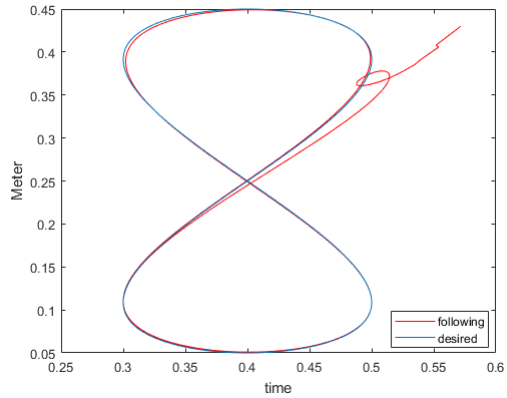
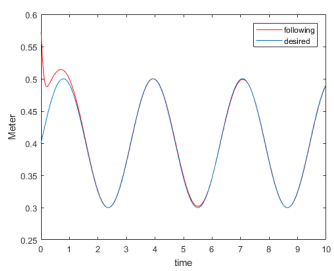
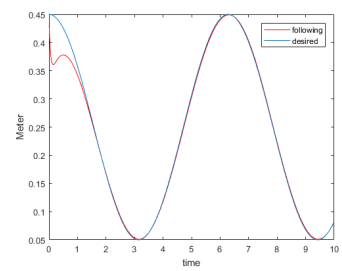


Figure 7: Desired and Following Trajectory



(a) End Effector following and desired trajectory in X coordinate



(b) End Effector following and desired trajectory in Y corrdinate

Figure 8: End Effector following trajectory

5.1 Results for $\beta = \sqrt{\det[(J^+)^T M^2 J^+]}$

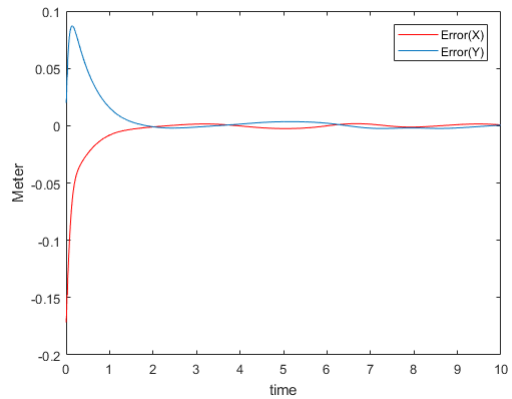


Figure 9: Error $x_d - x$ with respect to t

$y_a(q)$ and $\beta(q)$ functions are shown

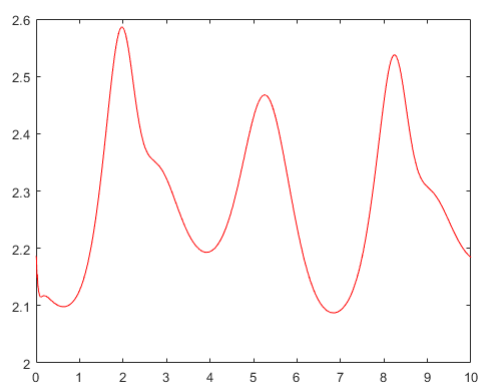


Figure 10: $\beta(q)$

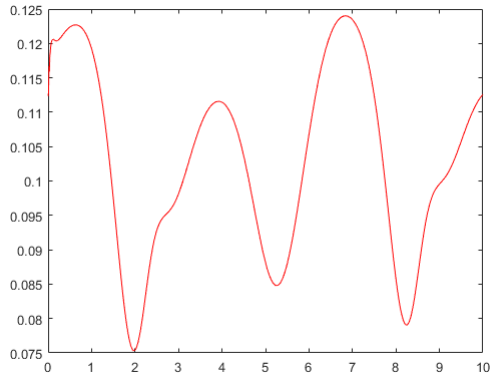


Figure 11: $y_a(q)$

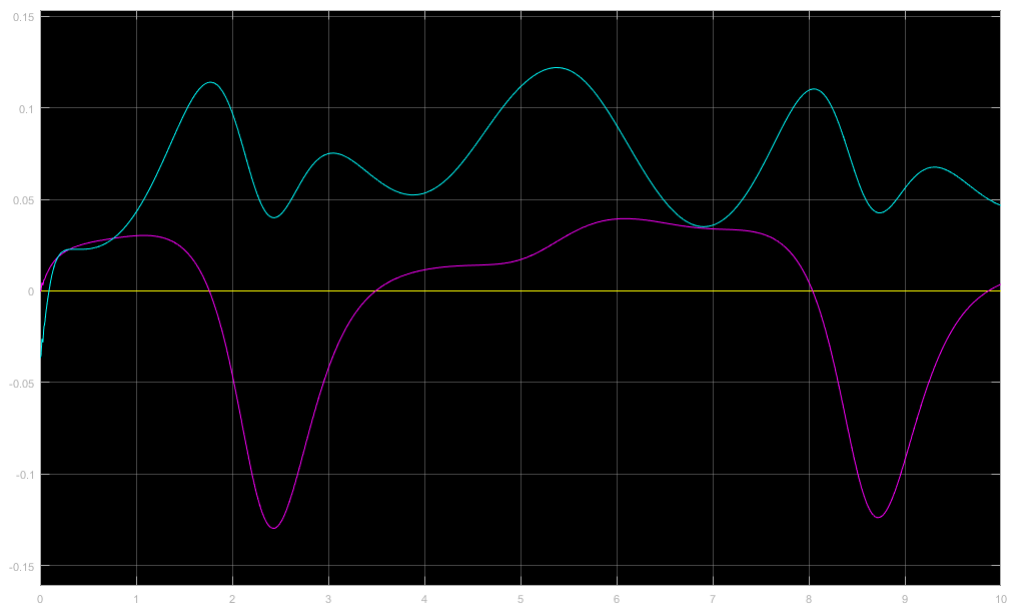
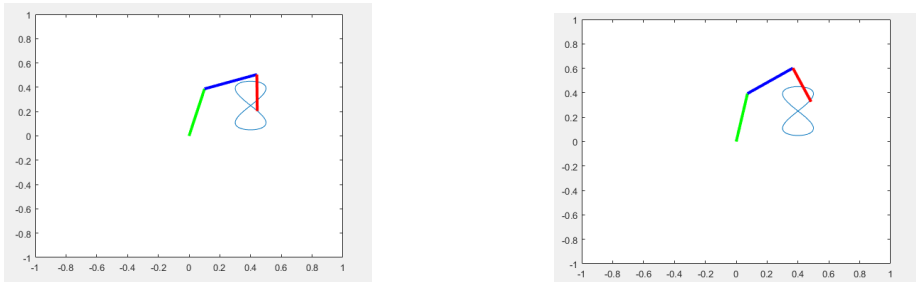


Figure 12: $J_s(q)$

As you can see in following figure, the robot perform its task as expected,



(a) Robot Position at $t = 1$ (b) Robot Position at $t = 2$

Figure 13: Animation of RRR

5.2 Results for $\beta = \sqrt{\det[(J^+)^T I^2 J^+]}$

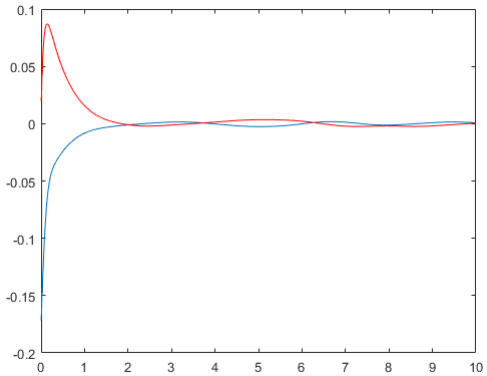


Figure 14: Error $x_d - x$ with respect to t

$y_a(q)$ and $\beta(q)$ functions are shown

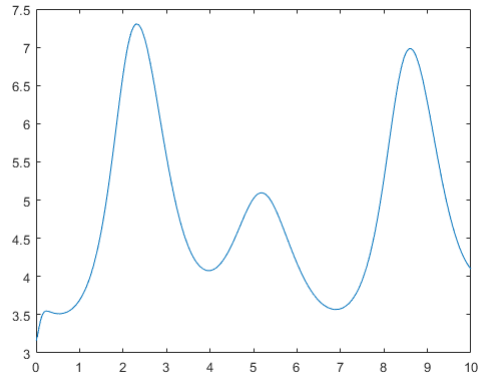


Figure 15: $\beta(q)$

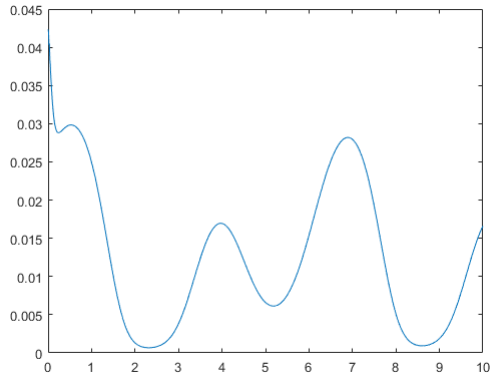


Figure 16: $y_a(q)$

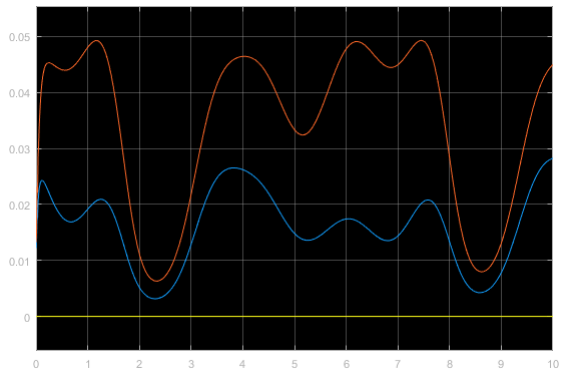
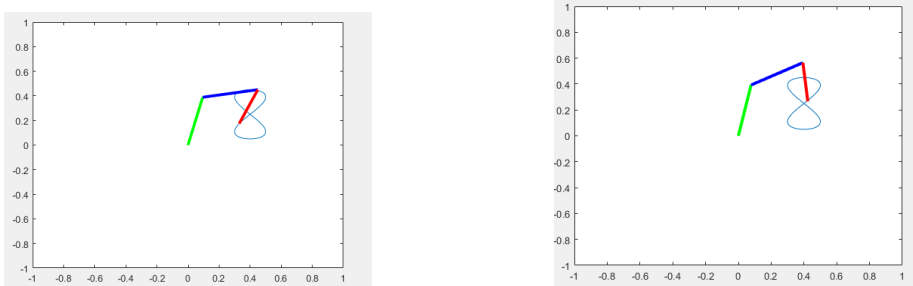


Figure 17: $J_s(q)$

As you can see in following figure, the robot perform its task as expected,



(a) Robot Position at $t = 1$

(b) Robot Position at $t = 2$

Figure 18: Animation of RRR

6 Conclusion

The objective of this project is to simulate redundant manipulator with $\beta(q)$ function mentioned in Section Beta Function which is Withstanding Impact.As can be seen in Figure 15 and 16 , While $\beta(q)$ is increasing , $y_a(q)$ is decreasing as anticipated. And Figure 14 illustrates that Error $(x_d - x)$ goes to the zero.

References

[1] Tatlicioglu, Enver *Control of Nonlinear Mechatronic Systems*, Clemson University, 2007.

[2] Walker, Ian D *Impact Configurations and Measures For Kinematically Redundant and Multiple Armed Robot Systems*,